Options Pricing and Risk Engine: Reproducible Valuation, Sensitivity Analysis, and Self-Financing Delta Hedging

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Abstract

This report presents a reproducible Python implementation of vanilla option valuation and risk analysis under constant-parameter Black–Scholes–Merton dynamics with continuous dividends. European analytic prices and Greeks provide controls for independent Cox–Ross–Rubinstein trees and Crank–Nicolson finite differences with Rannacher startup. American exercise is treated as an obstacle problem solved by projected successive over-relaxation, with complementarity diagnostics and mesh-estimated exercise boundaries. Numerical sensitivity studies separate perturbation size from spatial and temporal refinement and retain warnings near interpolation and exercise transitions. Signed portfolio scenarios account explicitly for expiry settlement, cash financing and local Greek P&L approximations. Deterministic hedge demonstrations compare unhedged, static and rebalanced Delta strategies using an auditable self-financing stock and cash ledger. Automated tests and synthetic experiments accompany tables, figures and execution provenance. Results are benchmark-specific; grid boundaries, American sensitivities and individual-path hedge outcomes are not presented as universal accuracy or performance guarantees.

Scope, provenance and rights

Software version: **1.0.0**. This synthesis uses the checked-in synthetic experiments generated with **1.0.0rc1**, retaining their original source hashes, commit references, warnings and timings. It does not relabel those measurements as new experiments. The historical implementation reference is `68aaee42a115667068495f8df7bb2e3d964c9bb8`; release preparation and its full local validation are recorded at `5732ea3c8730679f9e64b8cd4ca871924a36f0b8`. Subsequent publication changes do not alter the financial algorithms.

The software is licensed under MIT. This report and its original figures are licensed under **CC BY-SA 4.0**, Copyright (c) 2026 Matheus de Oliveira Gomes. Terms: https://creativecommons.org/licenses/by-sa/4.0/. **Zenodo DOI: pending deposit.** No deposit or DOI is claimed in this document. The software release is identified by the repository tag `v1.0.0` at https://github.com/diracadjoint/quant-options-pricing.

The engine covers one underlying and one currency with constant annualized volatility, continuously compounded interest and continuous dividend yield. Spot and strike are strictly positive, remaining maturity and volatility are nonnegative, and all numeric inputs must be finite. Times are years; rates, yields and volatility are decimals. Runtime pricing uses the Python standard library. Plotting dependencies are optional. There are no market observations, calibration, transaction costs, stock-borrow costs, margin or date conventions.

Model and independent valuation methods

Under risk-neutral dynamics,

$$dS = (r-q)S\,dt + \sigma S\,dW, \qquad \mathcal{L}V = \tfrac{1}{2}\sigma^2 S^2 V_{SS} + (r-q)SV_S - rV.$$

For European claims with remaining maturity τ, the engine implements

$$d_1 = \frac{\log(S/K) + (r - q + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau},$$
$$C = Se^{-q\tau}N(d_1) - Ke^{-r\tau}N(d_2), \qquad P = Ke^{-r\tau}N(-d_2) - Se^{-q\tau}N(-d_1).$$

The normal CDF uses `erfc`. Expiry returns intrinsic payoff, and zero volatility returns the positive part of the discounted deterministic spot-strike difference. No arbitrary epsilon replaces a degenerate input. Analytic Greeks are unavailable at zero maturity or volatility rather than silently replaced with zero.

CRR uses reciprocal up/down factors and validates the risk-neutral probability without clipping it. American rollback compares continuation against payoff. Even and odd step counts are studied separately. The finest pair is an approximate reference, not an exact American price.

The finite-difference engine uses a uniform spot grid, Crank–Nicolson time integration and four implicit half steps replacing the first two full steps. Europeans use tridiagonal elimination. Americans solve the obstacle

$$V - g \geq 0, \qquad V_\tau - \mathcal{L}V \geq 0, \qquad (V-g)(V_\tau - \mathcal{L}V) = 0$$

with projected successive over-relaxation. Both update and complementarity criteria must converge; failures raise exceptions. Drift upwinding is applied locally when needed for the required spatial sign structure. A small algebraic residual does not bound domain, discretization or interpolation error. The PDE currently requires $r, q \geq 0$; BSM and CRR accept negative rates/yields.

Price, domain and exercise-contact validation

The European control at $S = K = 100$, $\tau = 1$, $r = .05$, $q = 0$, $\sigma = .20$ is call **10.450583572185565**, put **5.573526022256971**. Tests include put–call parity, discounted bounds, degenerate cases and qualitative monotonicity. Spatial, temporal and upper-domain refinement are varied separately.

For the American benchmarks, $S = K = 100$, $\tau = 1$, $\sigma = .20$, $S_{max} = 400$, $M = 640$ and $N_t = 200$:

Claim	r	q	CN/PSOR	CRR2400/2401 average
Put	.05	0	6.08883942	6.09051979
Call	.02	.10	5.22387677	5.22533078

These are historical retained values. Core tests require absolute differences below .003 against each finest CRR parity for the specified fine-grid cases, not universal accuracy over all model inputs. Domain expansion at fixed local spacing includes both puts and dividend calls. The exercise frontier uses positive-payoff grid contacts and is quantized by mesh spacing. Frontier stability, value matching and continuation-side slopes have separate checks; price convergence alone does not certify a free boundary.

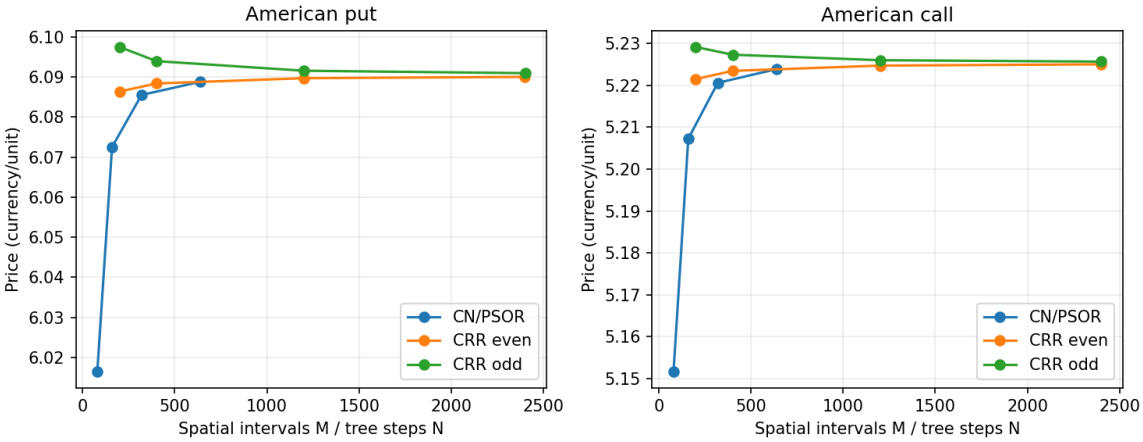


Figure 1: Historical American price refinement, with separate CRR parities.

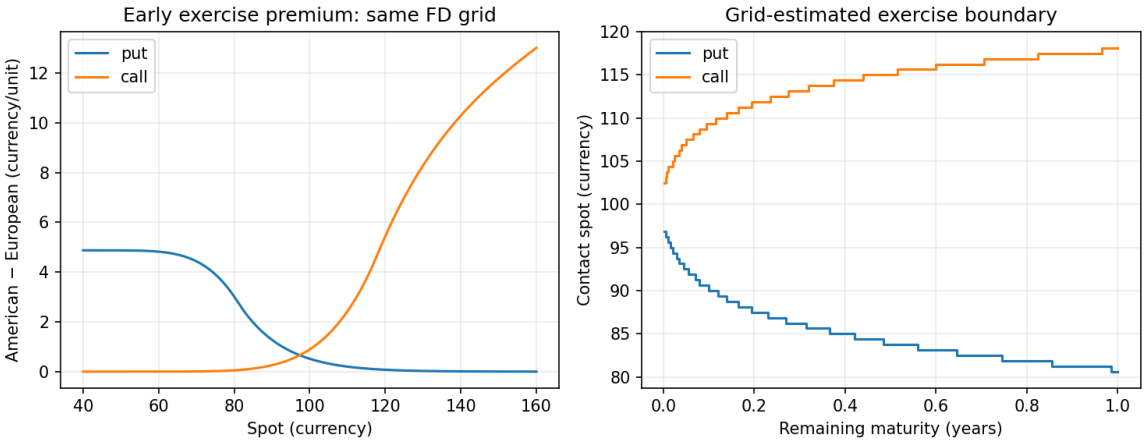


Figure 2: Historical exercise premium and mesh-estimated contacts.

Numerical Greeks and portfolio scenarios

Delta, Gamma, Vega, Rho and calendar Theta are computed with explicit derivative units. Vega/Rho are per absolute unit, and calendar Theta is per year. American bumps reprice exercise rights. Second-order central stencils are preferred; documented forward stencils handle parameter boundaries. Unavailable values, solver diagnostics and exercise-regime warnings remain in returned results.

Bump sensitivity is studied at fixed domain and mesh; spatial and temporal sensitivity are studied at fixed bumps while varying one resolution at a time. Linear price interpolation can suppress or amplify Gamma when the spot bump is smaller than grid spacing. Near exercise transitions, reducing a bump is not automatically an improvement and no universal smooth Gamma is claimed.

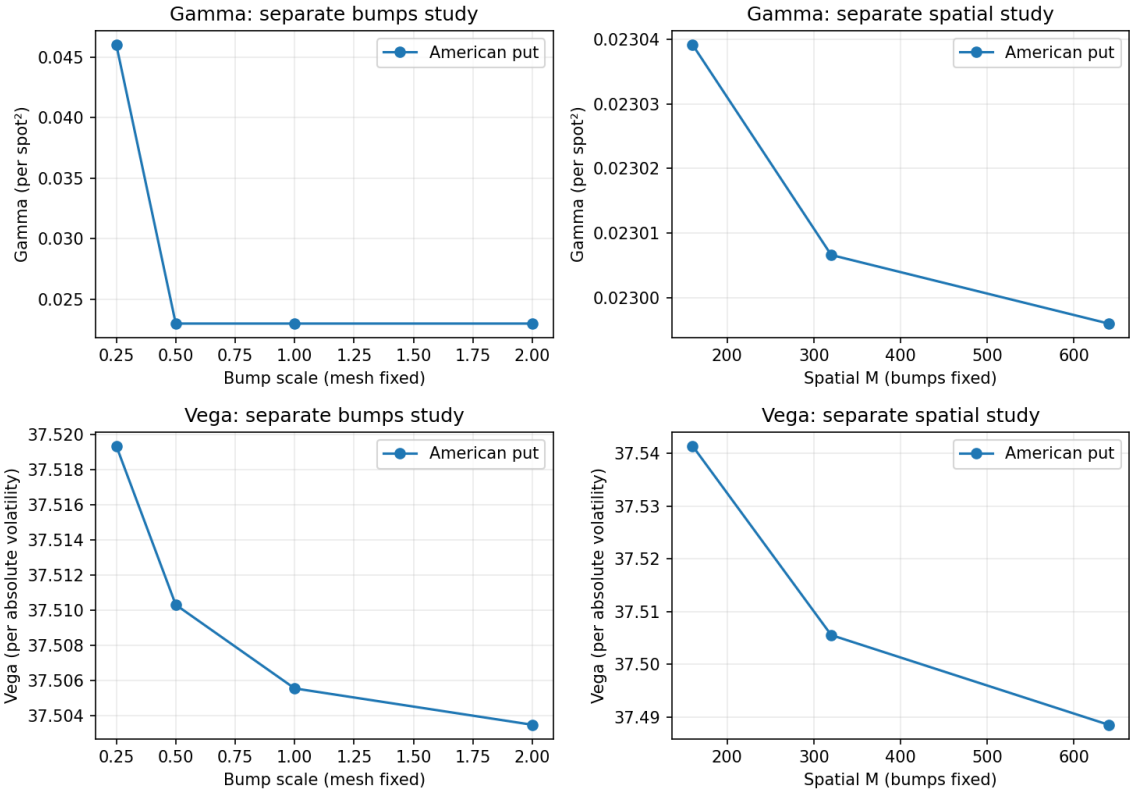


Figure 3: Historical bump and spatial-mesh sensitivity, including flagged interpolation behavior.

Portfolio value and each available Greek scale with signed quantity times positive contract multiplier. Scenario shocks occur immediately and then hold spot flat through the horizon. Expired options settle once into cash; each settlement accrues from its expiry at the scenario rate. Already-due positions settle at the original spot before shocks. Positive and negative cash use the same rate. Local P&L includes Delta, Vega, Rho and Theta; the second approximation adds only $\Gamma(\Delta S)^2/2$, omitting cross terms and other second derivatives. On crossing a live leg's expiry, the local Taylor estimate is explicitly unavailable while fully repriced settlement P&L remains defined.

Self-financing Delta hedge accounting

Paths are deterministic and piecewise linear between supplied increasing times. The desired stock inventory is $H = -\Delta_{\Pi}$, with

$$W = \Pi + HS + B, \qquad B_{after} = B_{before} - (H_{new} - H_{old})S.$$

Opening cash finances option value and stock against an explicitly chosen initial wealth. Trades preserve wealth up to floating-point rounding. Between events, stock units stay fixed and cash compounds continuously. Nominal stock dividends use $qH(S_{left} + S_{right})\Delta t/2$, credited at the interval end. This is exact for the nominal dividend integral along a linear segment, but omits intra-interval interest on dividends. Negative stock pays dividends. At expiry the option pays cash once, and the hedge closes at the last expiry or final horizon. Shorter horizons retain live option marks and an explicit warning.

The ledger separates option P&L, stock P&L, financing and dividends. Unavailable Delta raises; warned Delta raises by default or must be explicitly allowed and marked. American observations retain optimally priced exercise rights, but the demonstration **holds options to expiry without early exercise or assignment**.

The short European call example has quantity -1, multiplier 100, $K = 100$, $\tau = 1$, $\sigma = .20$, $r = .05$, $q = .02$, initial wealth 1000 and 65 observations. Historical final wealth is:

Path	Unhedged	Static	Dynamic every observation
Rise	-478.72048394	822.74245517	1338.92099701
Fall	2021.27951606	358.66054256	1315.64159935
Reversal	2021.27951606	1855.73909137	1562.73426517

Risk-free capitalized wealth is **1051.27109638**. These stress paths do not reproduce diffusion quadratic variation. Higher rebalance frequency need not improve every individual path or dominate static/unhedged positions.

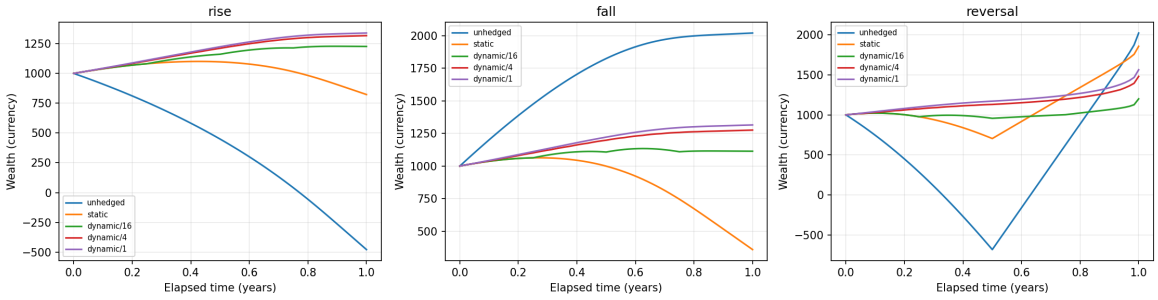


Figure 4: Historical wealth comparison under common paths and observation grids.

Reproduction, executed checks and limitations

From a repository checkout, with Python 3.12 for the plotting lock:

```
python -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements/visualization-py312.txt
python -m pip install --no-build-isolation .
python -m pytest -q
python examples/reproduce.py --mode full --output work/report-results
python scripts/verify_results.py work/report-results
python scripts/check_report_tables.py work/report-results
```

The release-preparation check on Python 3.12.14 executed all **802 tests** in 303.23 seconds. A wheel installed outside the checkout reported version 1.0.0 and the approved MIT metadata/license file. Historical artifact, accounting, reference-value and five-report table checks passed. Fast CI passed on Python 3.10, 3.12 and 3.13. New release workflow outcomes are separate observations and are recorded in GitHub Actions; this report does not preclaim their completion.

Historical artifacts comprise 32 outputs plus a manifest, including nine CSVs and eight PNG/SVG figure families. The manifest records configuration, source hashes, versions, environment and individual wall-time samples. New runs use separate output directories. Wall timings are not isolated benchmarks or speedup claims. Machine-dependent timing columns are excluded from report-value checks.

Finite floating-point range, tail underflow and subtractive cancellation remain limitations. Upper PDE boundaries are approximations; exercise contacts and bumped Greeks need refinement assessment. The implementation does not model discrete dividends, American assignment, transaction costs, margin, funding spreads, multi-asset/currency portfolios, stochastic volatility or market data. The numerical budgets and deterministic path outcomes are benchmark-specific.

References

- Black, F. and Scholes, M. (1973). The Pricing of Options and Corporate Liabilities. *Journal of Political Economy*, 81(3), 637–654.
- Merton, R. C. (1973). Theory of Rational Option Pricing. *Bell Journal of Economics and Management Science*, 4(1), 141–183.
- Cox, J. C., Ross, S. A. and Rubinstein, M. (1979). Option Pricing: A Simplified Approach. *Journal of Financial Economics*, 7(3), 229–263.
- Repository source, detailed validation reports, artifact data and provenance: https://github.com/diracadjoint/quant-options-pricing/tree/v1.0.0.